

- 1 (a) (i) acceleration (allow a definition of acceleration)..... B1
- (ii) the velocity is decreasing or force/acceleration is in negative direction – accept ‘body is decelerating’/‘slowing down’ B1 [2]
- (b) (i) e.g. separation of dots becomes constant/does not continue to increase (must make a reference to the diagram) B1
- (ii)1 distance = 132 cm..... B1
- (ii)2 at constant speed, distance travelled in 0.1 s = 25 cm
(allow ± 1 cm)..... C1
distance = 132 + (4 x 25)
= 232 cm A1 [4]
- (c) $s = ut + \frac{1}{2}at^2$
 $1.6 = \frac{1}{2} \times 9.8 \times t^2$ (allow $g = 10 \text{ m s}^{-2}$ C1
 $t = 0.57 \text{ s}$ C1
hence 6 photographs (‘bald’ answer scores 2 marks only)..... A1 [3]

2 (a) mass: measure of body’s resistance/inertia to changes in